

Web Application Development

IS4102

Lecture Note 08-Semantic Web Techniques

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Web 3.0 -the semantic
Web - is about the meaning
of data.

This is a Film

This released in 1800 theaters in India

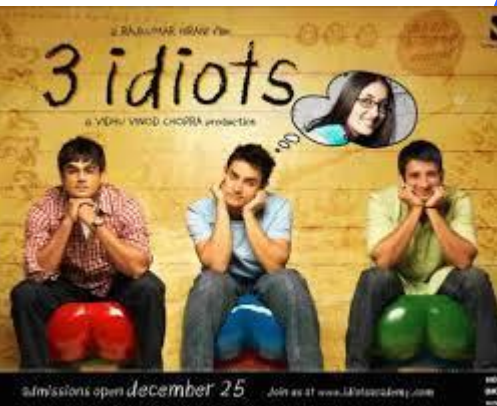
Main actor of the film is Amir Khan

This film is from the India

This is released in 2009

The picture on the CD cover shows the main actors

This film is directed by RajkumarHirani





Why do we want to add meaning to data ?

When a computer understands what data means, it can do intelligent search, reasoning and combining

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Meaning is about understanding. To understand we need a language. A language starts with words.

Things mean something in words Online, we describe things with,

XML.

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This is film collection

The first film is “Titanic” and is from USA. It was directed by James Cameron.

The second film is “chennai express” and is from India. It was directed by Rohit Shetty.

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```
<?xml version="1.0" encoding="ISO-8859-1"?>
```

```
<collection name="film collection">
```

```
<film>
```

```
<title>Titani</title>
```

```
<country>USA</country>
```

```
<director>James Cameron</director>
```

```
</film>
```

```
<film>
```

```
<title>Chennai Express</title>
```

```
<country>India</country>
```

```
<director>Rohit Shetty</director>
```

```
</film>
```

```
</collection>
```

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We can't understand words alone. We also need grammar. Online grammar is **RDF (Resource Description Framework)**

.

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subject

predicate



object

This film is from Sri Lanka.

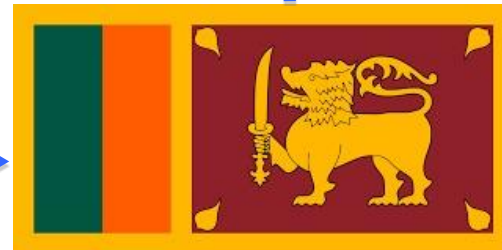
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With **RDF Scheme**, we can define concepts and make simpler relations between them.

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This film is from Sri Lanka, *hence* from South Asia.



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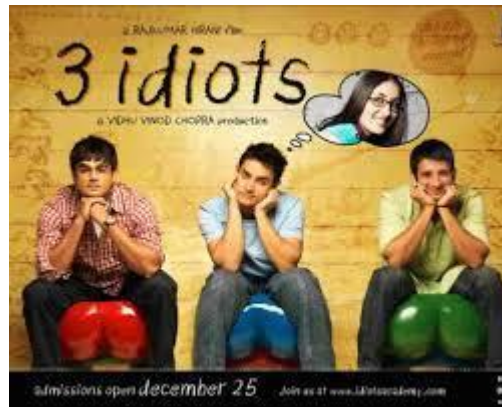


But, RDF scheme is limited. A language needs more **expression** and **logic** to make good reasoning possible.

That's why **OWL**(**The Web Ontology Language**) was invented.

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Finally, to reason we need rules.



I got this film from my **uncle**.

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I $\xrightarrow{\text{son of}}$ mother $\xrightarrow{\text{brother}}$



The rule for calling someone my uncle is that one of my parents has a brother.

Rules are formulated in **SWRL**(**Semantic Web Rule Language**).

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```
<ruleml:imp>
  <ruleml:_rlab ruleml:href="#example1"/>
  <ruleml:_body>
    <swrlx:individualPropertyAtom swrlx:pr
operty="hasParent">
      <ruleml:var>x1</ruleml:var>
    </swrlx:individualPropertyAtom>
    <swrlx:individualPropertyAtom swrlx:pr
operty="hasBrother">
      <ruleml:var>x2</ruleml:var>
      <ruleml:var>x3</ruleml:var>
    </swrlx:individualPropertyAtom>
  </ruleml:_body>
  <ruleml:_head>
    <swrlx:individualPropertyAtom swrlx:pr
operty="hasUncle">
      <ruleml:var>x1</ruleml:var>
    </swrlx:individualPropertyAtom>
  </ruleml:_head>
</ruleml:imp>
```



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Words in XML

Grammar in RDF (scheme) and OWL

Rules in SWRL